Musaddique Lalkot

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PROFESSIONAL SUMMARY

Generative AI Engineer specializing in **Agentic AI** and **multi-agent systems**, with hands-on experience designing, building and deploying production-grade, LLM-powered automation platforms for the **insurance (P&C) domain** – end-to-end **claims processing** and **commercial underwriting**. Skilled in **Python, FastAPI, LangGraph, LangChain, Azure OpenAI (GPT-4.1), Azure AI Foundry, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG)** and **vector databases**, with full-stack delivery using **React, TypeScript** and **Azure PostgreSQL**. Architected a **claims platform of 46 specialist agents and 4 orchestrators** across four business personas and a **16-phase underwriting pipeline** with **Human-in-the-Loop (HITL)** governance, **Guidewire** integration and containerized deployment on **Microsoft Azure**.

SKILLS

- **Programming Languages:** Python, TypeScript, JavaScript, SQL
- **Generative AI & LLMs:** Azure OpenAI (GPT-4.1, GPT-4o), Azure AI Foundry (Agents, Vector Stores, File Search), Agentic AI, Multi-Agent Orchestration, LangGraph, LangChain, Model Context Protocol (MCP), Retrieval-Augmented Generation (RAG), Prompt Engineering, Embeddings, Tool / Function Calling, Human-in-the-Loop (HITL)
- **Document AI & NLP:** Azure AI Document Intelligence, Intelligent Document Processing (IDP), Structured Data Extraction, Document Classification, Summarization, Speech-to-Text (Azure OpenAI GPT-4o Transcribe)
- **Vector Databases & Databases:** Azure AI Foundry Vector Stores, pgvector, ChromaDB, PostgreSQL (Azure Database for PostgreSQL), SQLite, Drizzle ORM
- **Backend Development:** FastAPI, Express.js, Node.js, RESTful APIs, Server-Sent Events (SSE) Streaming, Webhooks, Microservices Architecture, Event-Driven Architecture
- **Frontend Development:** React, Vite, Tailwind CSS, Shadcn UI, TanStack Query
- **Cloud & DevOps:** Microsoft Azure (Blob Storage, Container Instances, Container Registry, Key Vault, VNET), Docker, Azure DevOps CI/CD Pipelines
- **LLMOps & Observability:** Arize Phoenix (Prompt Management), LLM Evaluation, Logging & Monitoring
- **Integrations & Domain:** Guidewire PolicyCenter & ClaimCenter APIs, Gmail API (OAuth 2.0), Sanctions-List Screening, Insurance – FNOL, Claims Lifecycle, Underwriting, Fraud Detection (SIU)
- **Tools & Methodologies:** Git, GitHub, Azure DevOps, Agile/Scrum

WORK EXPERIENCE

Hexaware Technologies Limited Pune, India
[Software Engineer Level 1– Generative AI] *Present*

Agentic AI Claims Automation Platform (P&C Insurance) | Python, FastAPI, LangGraph, LangChain, MCP, Azure OpenAI, Azure PostgreSQL, Azure Blob Storage, React, TypeScript, Vite, Arize Phoenix, Guidewire

- Architected and developed an end-to-end **multi-agent AI platform** automating the complete **insurance claims lifecycle** – **FNOL** intake, triage, fraud detection, damage assessment, vendor assignment, reserve, settlement, payment and closure.
- Built **46 specialist AI agents and 4 orchestrator agents** using **LangGraph StateGraph** across **4 personas** (Policyholder, Adjuster, SIU, Vendor Manager), each served as a **FastAPI** microservice with **SSE streaming** chat endpoints.
- Designed **Model Context Protocol (MCP)** tool servers per persona using **FastAPI-MCP**, exposing business operations as LLM-callable tools consumed via **MultiServerMCPClient**; persona **orchestrator agents** aggregate **50+ tools** to drive the full claim journey.
- Implemented **Human-in-the-Loop (HITL) approval gates** across the Adjuster, SIU and Vendor orchestrators – code-enforced coverage and SIU-investigation reviews, audit-only triage and vendor-assignment gates, and adjuster-decided reserve and settlement gates – ensuring **responsible AI** and regulatory compliance.
- Extended the FNOL agents to support **multiple lines of business (Motor and Homeowners)** with a configuration-routed **FNOL Orchestrator** that selects each tool server through environment variables, so ported sub-apps move between servers without code changes.
- Delivered **voice and text FNOL intake** using **browser speech capture** (server-side diarized transcription in the standalone intake agent), plus **straight-through processing (STP) scoring**, duplicate-claim detection, policy coverage verification and claim-readiness checks.

- Integrated **Guidewire PolicyCenter and ClaimCenter REST APIs** for real-time Homeowners policy lookup and coverage validation (Motor uses a local policy store); stored claim evidence in **Azure Blob Storage** and claim state in a shared **Azure PostgreSQL** database.
- Built **SIU fraud-detection** agents (fraud risk scoring, network and entity-relationship analysis, behavioral analytics, watchlist updates) and **vendor-management** agents (matching, dispatch, ETA prediction, SLA compliance, cost benchmarking).
- Developed the **React 19 / Vite / TypeScript / Tailwind** claims portal with role-based dashboards for policyholders, adjusters, SIU investigators and vendor managers, including AI-driven **repair-vs-replacement** recommendations with adjuster override logic.
- Enabled **LLMOps** with **Arize Phoenix** for centralized **prompt management**.
- **Impact:** [Add measured outcome – e.g. X% reduction in FNOL handling time, X% of claims processed straight-through].

AI-Powered Commercial Underwriting Automation (Azure AI Foundry) | Python, LangGraph, FastAPI, MCP, Azure AI Foundry, Azure OpenAI GPT-4.1, Azure AI Document Intelligence, pgvector, ChromaDB, Azure Blob Storage, PostgreSQL, React, Express.js, TypeScript, Docker, Azure Container Instances, Guidewire

- Developed an **agentic underwriting pipeline** automating the commercial insurance submission lifecycle across **16 phases** – from broker email ingestion to **Guidewire quote, bind and policy issuance**.
- Built a **LangGraph**-based **Email Intent Agent** orchestrating **24 MCP tool modules** (FastAPI + FastAPI-MCP) for email reading via **Gmail API**, message de-duplication, attachment download, **LOB detection**, **document layout classification**, **field extraction** and summarization.
- Engineered the **Intelligent Document Processing (IDP)** / **RAG** layer through three iterations – **ChromaDB** → **PostgreSQL pgvector** → **Azure AI Foundry vector stores** – moving to per-submission, cloud-native indexing with **incremental indexing** of new attachments and **File Search**-grounded extraction.
- Implemented **LOB- and document-type-specific extraction** driven by a prompt/attribute library in PostgreSQL, producing structured JSON with **confidence scores** for ACORD and non-ACORD documents.
- Automated **licensing and sanctions-list screening**, **perils data capture** (geocoding and earthquake APIs with a peril-risk lookup), **risk score calculation** and AI-assisted **accept/reject** recommendations, followed by quote PDF generation and broker email drafting.
- Designed **3 HITL gates** (submission review, eligibility, risk acceptance) with **stateless gate-resume endpoints** (graph halts after each gate; state in Blob and PostgreSQL) for pause/resume across underwriter approvals.
- Built the **Underwriter Workboard** (**React, TypeScript, Express.js, Drizzle ORM, Azure PostgreSQL**) with **event-driven ingestion** via Azure Blob monitors and webhooks, **confidence-based auto-approval (≥90%)**, a forward-only stage workflow and **RAG chat** over submission documents.
- Containerized services with **Docker** and deployed to **Azure Container Instances** inside a **VNET**, with secrets managed via **Azure Key Vault**; used **Arize Phoenix** for prompt versioning.
- **Impact:** [Add measured outcome – e.g. X% faster submission turnaround, X% reduction in manual data entry].

EDUCATION

MIT ADT - School of Engineering
B.Tech. - Computer Science & Engineering - MIT ADT - School of Engineering **2020 - 2024**
[CGPA 7.7 / 10]

Prior Experience

- IBM Skill Build- What is Artificial Intelligence? Successfully certified
- Internship – Web Development (TwiLearn EduTech Pvt Ltd)

ACHIEVEMENTS

- Employee of the Month Award
- Promoted to Software Engineer Level 1